



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Business Systems

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.Tech, II Computer Science and Business Systems

Course Code & Title: AL3451 & Machine Learning

Name of the Faculty member (s): Mrs.B.Yazhini AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit V /Measuring Classifier Performance

- **Course Outcome:** CO 5

- **Activity Chosen:** Virtual Lab

- **Justification:**

The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is an on screen simulator that learners use to test ideas and observe results. Since the principles of operators is mandatory for the programming language, hands on demonstration is given to the students through Virtual lab by using the following link

<https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/simulation.html>

- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**

The course instructor explained the concept of the various operators. He opened the Virtual Lab using aforementioned link and demonstrates the various operators working to the students. At last 10 minutes, the course instructor chosen few students and instructed them to explain the various examples of operators by the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5
CO5	3	3	3	2	3

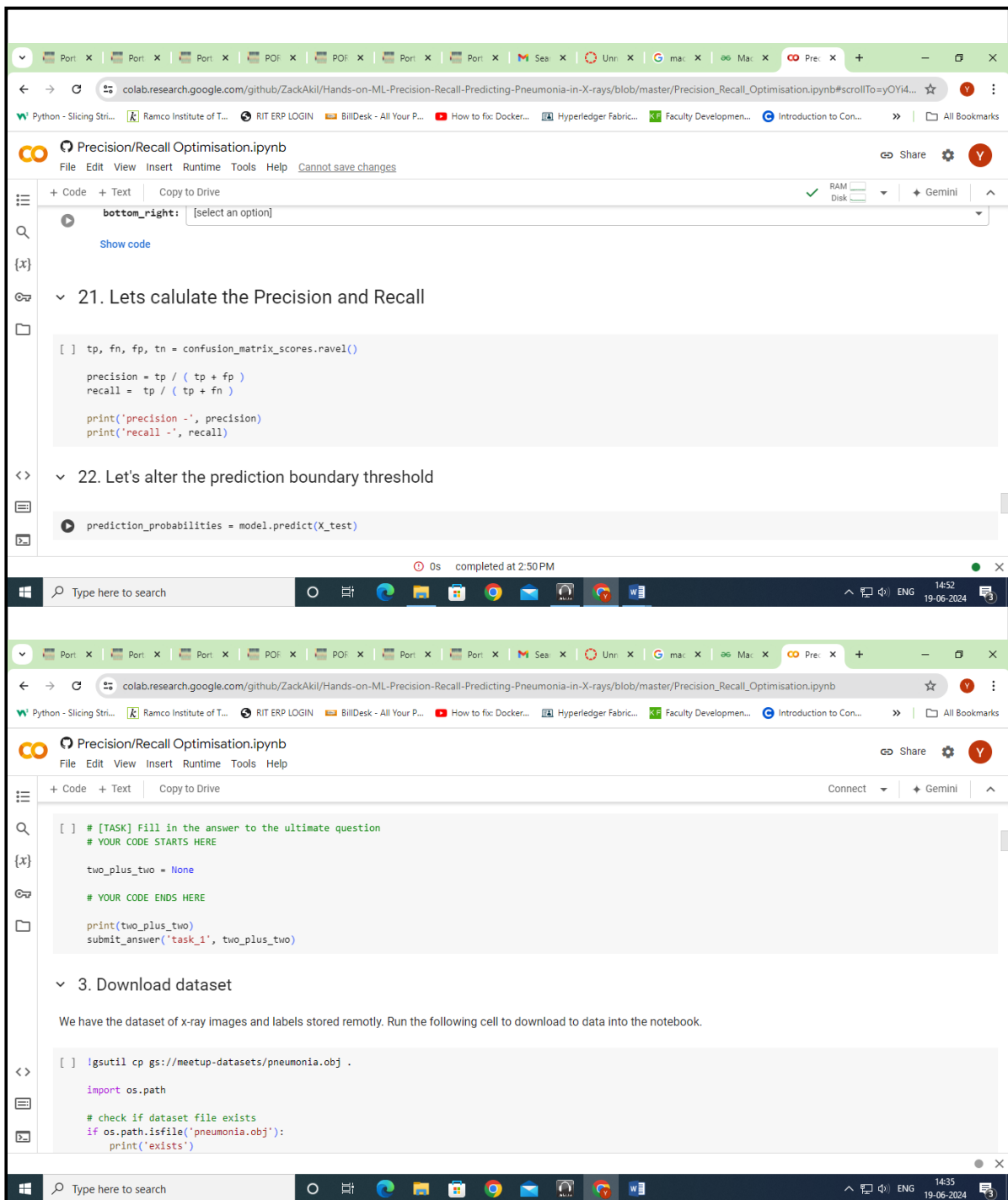
(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 1	PO 2	PO 3	PO4
	3	3	2	3
Justification for correlation	Students can be able to perform cross validation and resampling	Students will be able to analyze the performance of machine learning classifiers	Students will gain proficiency in analysis of machine learning experiments	Students will apply research-based knowledge and methodologies to validate the effectiveness and performance of various Machine learning algorithms
Innovative practice	PSO 2	PO 5		
	3	3		
Justification for correlation	To create, and apply the techniques of Machine learning to forecast future events in the domain of Healthcare, Education, and Agriculture, Manufacturing, Automation, Robotics, Transport, etc.,	Students will use modern tools and platforms for some machine learning experiments like K-fold CV, bootstrapping etc.		

- Images / Screenshot of the practice:

Fig 1: Demonstration Measuring Classifier Performance



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students easily understand the working of Data types and Loops and its corresponding code.
 - Students said that the pretest and posttest is very useful to recollect the concepts.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students had understood the concepts of the various Data types
 - Most of the students were actively participated and enjoyed once they answered correctly in pretest.

- ❖ ***Challenges faced in implementation:***

- Some students only involved in the demonstration activity.
 - The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VII_Vlab_Demo.pdf
 - ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>

Signature of Faculty Member

HOD